Integrals Chapter Notes

Grade 12 | Free | From first principles to advanced reasoning

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These notes develop integration from reversing differentiation to evaluating definite integrals efficiently. Start with Sections 1–8, then use Sections 9–14 to strengthen reasoning. The companion free set contains 50 questions with full solutions. Worked examples here are separate teaching examples.

Learning goals. Recognise a useful method, show valid substitutions, state domain restrictions, use definite-integral properties with their hypotheses, and check an answer without repeating the whole calculation.

Notation. All angles are measured in radians. $\ln x$ is the natural logarithm, $\sin^{-1} x$ denotes the inverse sine, and $\csc x = 1/\sin x$. The letter C denotes an arbitrary real constant. Unless otherwise stated, use exact values of π.

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1 What integration means

If $F'(x) = f(x)$, then F is an antiderivative of f. The notation

$$\int f(x)\,dx = F(x) + C$$

describes a family of functions. The symbol dx identifies the variable of integration. A definite integral $\int_a^b f(x)\,dx$ is a number representing accumulated signed change from a to b. It is not the same kind of object as an indefinite integral.

Why is C needed? Differentiating $F(x) + 7$ or $F(x) - 11$ gives the same derivative. On a connected interval, any two antiderivatives differ by a constant. If the domain has separate intervals, the constants on those intervals need not agree.

Example 1. A quantity has derivative $q'(x) = 8x - 3$ and $q(1) = 6$. Integration gives $q(x) = 4x^2 - 3x + C$. The initial value gives $6 = 4 - 3 + C$, so $C = 5$ and $q(x) = 4x^2 - 3x + 5$.

Before you begin. Revise the chain rule, product rule, factorisation, completing the square, and the identities $\sin^2 x + \cos^2 x = 1$, $1 + \tan^2 x = \sec^2 x$. Most integration methods reverse one of these familiar operations.

2 Basic antiderivatives and their conditions

For every formula below, work on an interval where the integrand and the displayed antiderivative are real and defined. Add C to each answer.

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1), \qquad \int \frac{dx}{x} = \ln|x| + C,$$

$$\int e^{kx}\,dx = \frac{e^{kx}}{k} + C \quad (k \neq 0), \qquad \int b^x\,dx = \frac{b^x}{\ln b} + C \quad (b > 0,\ b \neq 1),$$

$$\int \sin x\,dx = -\cos x + C, \qquad \int \cos x\,dx = \sin x + C,$$

$$\int \sec^2 x\,dx = \tan x + C, \qquad \int \csc^2 x\,dx = -\cot x + C,$$

$$\int \sec x \tan x\,dx = \sec x + C, \qquad \int \csc x \cot x\,dx = -\csc x + C,$$

$$\int \tan x\,dx = -\ln|\cos x| + C, \qquad \int \cot x\,dx = \ln|\sin x| + C,$$

$$\int \sec x\,dx = \ln|\sec x + \tan x| + C, \qquad \int \csc x\,dx = \ln|\csc x - \cot x| + C.$$

The power rule fails at $n = -1$: the logarithmic rule takes its place. For noninteger powers, the real domain needs attention. Linearity allows constants to be taken outside the integral and sums to be integrated term by term. There is no corresponding rule saying the integral of a product equals the product of the integrals.

Example 2. For $x > 0$,

$$\int \left(9x^2 + \frac{2}{\sqrt{x}} - \frac{7}{x}\right) dx = 3x^3 + 4\sqrt{x} - 7\ln x + C.$$

Check: differentiating the answer gives $9x^2 + 2/\sqrt{x} - 7/x$.

3 Substitution as the reverse chain rule

Look for a function together with its derivative. If $u = g(x)$, replace both the expression and the differential: $du = g'(x)dx$. Do not leave an unexplained mixture of u and x.

$$\int f(g(x))g'(x)\,dx = F(g(x)) + C \quad \text{when } F' = f.$$

Useful patterns are g'/g, $g'\,g^n$, $g'e^g$, and $g'/(a^2+g^2)$.

Example 3. Evaluate $\int (8x-3)(4x^2-3x+10)^6\,dx$. Set $u = 4x^2-3x+10$. Then $du = (8x-3)dx$, so the answer is $u^7/7 + C = (4x^2-3x+10)^7/7 + C$.

Example 4. Evaluate $\int e^{2x}/(7+e^{2x})\,dx$. Set $u = 7+e^{2x}$, $du = 2e^{2x}dx$. The factor $1/2$ gives $\frac{1}{2}\ln(7+e^{2x}) + C$.

Example 5. For $x > 0$, substitute $t = \sqrt{x}$ in $\int dx/(\sqrt{x}+3)$. Since $x = t^2$ and $dx = 2t\,dt$, divide $2t/(t+3) = 2 - 6/(t+3)$. The answer is $2\sqrt{x} - 6\ln(\sqrt{x}+3) + C$.

Self check. Differentiate $\frac{1}{5}\ln(2+e^{5x})$. The result is $e^{5x}/(2+e^{5x})$, showing exactly where the reciprocal inner derivative enters.

4 Quadratics and standard forms

For $a > 0$, the following formulas cover many algebraic integrals:

$$\begin{aligned}
\int \frac{dx}{a^2+x^2} &= \frac{1}{a}\tan^{-1}(x/a) + C,\\
\int \frac{dx}{a^2-x^2} &= \frac{1}{2a}\ln\left|\frac{a+x}{a-x}\right| + C,\\
\int \frac{dx}{x^2-a^2} &= \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C,\\
\int \frac{dx}{\sqrt{a^2-x^2}} &= \sin^{-1}(x/a) + C \quad (|x| < a),\\
\int \frac{dx}{\sqrt{x^2+a^2}} &= \ln|x + \sqrt{x^2+a^2}| + C,\\
\int \frac{dx}{\sqrt{x^2-a^2}} &= \ln|x + \sqrt{x^2-a^2}| + C \quad (|x| > a).
\end{aligned}$$

Absolute values are essential in logarithmic forms. An antiderivative across a pole is not a license to evaluate a definite integral through that pole.

Complete the square before deciding a method:

$$Ax^2 + Bx + D = A\left(x + \frac{B}{2A}\right)^2 + D - \frac{B^2}{4A}, \qquad A \neq 0.$$

If a numerator is linear, express it as a multiple of the derivative of the quadratic plus a constant. This separates a logarithmic part from a standard-form part.

Example 6. Evaluate $\int (6x+13)/(x^2+4x+8)\,dx$. Since $6x+13 = 3(2x+4)+1$ and $x^2+4x+8 = (x+2)^2+4$,

$$I = 3\ln(x^2+4x+8) + \frac{1}{2}\tan^{-1}\left(\frac{x+2}{2}\right) + C.$$

Example 7. Evaluate $\int (4x+9)/\sqrt{x^2+4x+8}\,dx$. Split $4x+9 = 2(2x+4)+1$. The first part is $4\sqrt{x^2+4x+8}$. The second is $\ln|x+2+\sqrt{x^2+4x+8}|$. Add them and C.

For radicals in the numerator, use these forms or derive them by parts:

$$\int \sqrt{a^2-x^2}\,dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}(x/a) + C,$$
$$\int \sqrt{x^2+a^2}\,dx = \frac{x}{2}\sqrt{x^2+a^2} + \frac{a^2}{2}\ln|x+\sqrt{x^2+a^2}| + C,$$
$$\int \sqrt{x^2-a^2}\,dx = \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\ln|x+\sqrt{x^2-a^2}| + C.$$

Example 8. In $\int \sqrt{12+4x-x^2}\,dx$, write the root as $\sqrt{16-(x-2)^2}$. Let $u = x-2$ and apply the first formula with $a = 4$. The answer is

$$\frac{x-2}{2}\sqrt{16-(x-2)^2} + 8\sin^{-1}\left(\frac{x-2}{4}\right) + C$$

on $-2 < x < 6$. This illustrates how a general quadratic radical reduces to a standard form.

5 Rational functions and partial fractions

If the numerator degree is at least the denominator degree, divide first. Next factor the denominator over the reals. For distinct linear factors use $A/(x-a) + B/(x-b)$; for a repeated factor include every power, for example $A/(x-a) + B/(x-a)^2$. For an irreducible quadratic use a linear numerator $(Bx+D)/(x^2+px+q)$.

Example 9. Decompose $\frac{4x+11}{(x+1)(x+4)} = \frac{A}{x+1} + \frac{B}{x+4}$. Clearing denominators gives $4x+11 = A(x+4) + B(x+1)$, hence $A = 7/3$, $B = 5/3$. The integral is $\frac{7}{3}\ln|x+1| + \frac{5}{3}\ln|x+4| + C$.

Example 10. Since $x+4 = (x+2)+2$,

$$\int \frac{x+4}{(x+2)^2}\,dx = \int\left(\frac{1}{x+2} + \frac{2}{(x+2)^2}\right)dx = \ln|x+2| - \frac{2}{x+2} + C.$$

Example 11. For $\int dx/[x(x^2+4)]$, write the fraction as $1/(4x) - x/[4(x^2+4)]$. Integrate to obtain $\frac{1}{4}\ln|x| - \frac{1}{8}\ln(x^2+4) + C$. Verify by combining the derivatives over a common denominator.

6 Integration by parts

The product rule gives $d(uv) = u\,dv + v\,du$, so

$$\int u\,dv = uv - \int v\,du.$$

Choose u so that differentiating it simplifies the problem and choose dv so that it can be integrated. A logarithm, inverse trigonometric function, or polynomial often makes a useful u, but this is a guide, not a compulsory ordering rule.

Example 12. For $\int x\cos 5x\,dx$, choose $u = x$, $dv = \cos 5x\,dx$. Then $v = \sin 5x/5$ and

$$I = \frac{x\sin 5x}{5} - \frac{1}{5}\int \sin 5x\,dx = \frac{x\sin 5x}{5} + \frac{\cos 5x}{25} + C.$$

Example 13. For $\int \tan^{-1}(3x)\,dx$, take $u = \tan^{-1}(3x)$ and $dv = dx$. This gives

$$I = x\tan^{-1}(3x) - \int \frac{3x}{1+9x^2}\,dx = x\tan^{-1}(3x) - \frac{1}{6}\ln(1+9x^2) + C.$$

Example 14. Let $I = \int e^x\cos 2x\,dx$ and $J = \int e^x\sin 2x\,dx$. Two uses of parts give $I = e^x\cos 2x + 2J$ and $J = e^x\sin 2x - 2I$. Substitute the second into the first to get $5I = e^x(\cos 2x + 2\sin 2x)$. Thus $I = e^x(\cos 2x + 2\sin 2x)/5 + C$. The repeated integral is an equation to solve, not a reason to restart.

7 Trigonometric integrals

For an odd power of sine, reserve one $\sin x$ and replace the remaining $\sin^2 x$ by $1-\cos^2 x$; then put $u=\cos x$. For an odd power of cosine, reserve one $\cos x$ and put $u=\sin x$. For two even powers, use half-angle identities. For a product at different frequencies, product-to-sum identities may be shorter.

Example 15. Evaluate $\int \sin^3 x\cos^2 x\,dx$. Write it as $\int(1-\cos^2 x)\cos^2 x\sin x\,dx$. Set $u=\cos x$, $du=-\sin x\,dx$. The answer is $-u^3/3+u^5/5+C=-\cos^3 x/3+\cos^5 x/5+C$.

Example 16. Since $\sin^2 x\cos^2 x=(1-\cos 4x)/8$,

$$\int \sin^2 x\cos^2 x\,dx = x/8-\sin 4x/32+C.$$

For radicals, substitutions $x=a\sin\theta$ in $\sqrt{a^2-x^2}$ and $x=a\tan\theta$ in $\sqrt{a^2+x^2}$ are useful. Choose the range of θ so that square-root signs are handled correctly; $\sqrt{\cos^2\theta}=|\cos\theta|$.

8 Definite integrals and the fundamental theorem

Partition $[a,b]$ into n equal parts of width $\Delta x=(b-a)/n$. For continuous f, the right-endpoint sums converge to

$$\int_a^b f(x)\,dx=\lim_{n\to\infty}\frac{b-a}{n}\sum_{r=1}^{n} f\left(a+\frac{r(b-a)}{n}\right).$$

Positive heights contribute positively and negative heights negatively. Consequently, signed integral and geometric area need not agree. Total area between a graph and the axis is $\int_a^b |f(x)|dx$.

If $F'=f$ and f is continuous on $[a,b]$, the fundamental theorem states

$$\int_a^b f(x)\,dx=F(b)-F(a).$$

Do not add C to a numerical definite-integral answer: the constants cancel. A substitution changes bounds: $u=g(a)$ to $g(b)$, with its differential and orientation retained. If instead you return to x before evaluation, use the original bounds.

Example 17. Evaluate $\int_0^2(4x+2)e^{2x^2+2x}\,dx$. Set $u=2x^2+2x$, so $du=(4x+2)dx$ and the bounds are $0, 12$. The answer is $[e^u]_0^{12}=e^{12}-1$.

Example 18. $\int_{-1}^2(x-1)dx=-3/2$, although the total area is positive. Split at $x=1$: the negative triangle has magnitude 2 and the positive triangle magnitude $1/2$. Signed integral is $-2+1/2=-3/2$; total area is $2+1/2=5/2$.

A boundary warning. $\int_{-1}^1 dx/x^2$ cannot be evaluated by simply subtracting the values of $-1/x$: the integrand has a pole at zero. The two one-sided improper integrals diverge. Check the interval before using the theorem.

9 Properties that shorten definite integrals

Assume all displayed integrals exist; for inequalities below take $a<b$.

$$\int_a^a f(x)dx=0, \qquad \int_b^a f(x)dx=-\int_a^b f(x)dx,$$

$$\int_a^b f(x)dx=\int_a^c f(x)dx+\int_c^b f(x)dx, \qquad \int_a^b f(x)dx=\int_a^b f(a+b-x)dx.$$

The last identity follows by substituting $u = a + b - x$. It reflects the interval about its midpoint. If $f(x) + f(a + b - x) = K$, addition gives $\int_a^b f = K(b - a)/2$.

On $[-a, a]$, an integrable odd function has integral zero; an even function has integral twice its integral from 0 to a. These rules require both the symmetry and existence of the integral. If $f(a+b-x) = f(x)$, then

$$\int_a^b xf(x)dx = \frac{a+b}{2}\int_a^b f(x)dx.$$

To derive this, reflect x to get a second expression with factor $a + b - x$ and add. Do not assume the identity for an arbitrary weight f.

Example 19. Find $I = \int_1^4 \frac{e^x}{e^x+e^{5-x}}dx$. Reflection replaces the numerator by e^{5-x}. Therefore $2I = \int_1^4 1\,dx = 3$, so $I = 3/2$.

Example 20. In $\int_{-3}^3[x^7/(2+x^2)+4x^2]dx$, the first term integrates to zero by oddness. The remaining even part is $8\int_0^3 x^2dx = 72$.

If f is integrable and periodic with period $T > 0$, then $\int_c^{c+T} f = \int_0^T f$ for every c. Split a shifted period at T and translate the remaining part to prove it. Thus $\int_c^{c+nT} f = n\int_0^T f$ for positive integers n.

Example 21. $|\sin 2x|$ has period $\pi/2$, and its integral over one period is 1. The interval from $\pi/7$ to $\pi/7 + 3\pi$ has six full periods. Its integral is therefore 6.

10 Modulus and piecewise functions

Find every point where the formula changes, split there, and use one valid expression on each subinterval. For $|g(x)|$, solve $g(x) = 0$ and test the sign on each piece. For $\lfloor g(x)\rfloor$, locate the relevant integer crossings. Changing finitely many function values does not change a Riemann integral when the function remains integrable.

Example 22. In $\int_0^4 |3x - 4|dx$, split at $4/3$. The two positive triangular areas are $(1/2)(4/3)(4) = 8/3$ and $(1/2)(8/3)(8) = 32/3$. The answer is $40/3$. Direct integration of $4 - 3x$ and $3x - 4$ gives the same result.

Example 23. For $\int_0^2\lfloor 3x\rfloor dx$, the values $0, 1, 2, 3, 4, 5$ each occupy width $1/3$. Hence the integral is $(0 + 1 + 2 + 3 + 4 + 5)/3 = 5$.

11 Integrals with variable endpoints

For continuous f, let $A(x) = \int_a^x f(t)dt$. Then $A'(x) = f(x)$. The symbol t is a dummy variable; x remains the free variable. Combining this with the chain rule gives

$$\frac{d}{dx}\int_{u(x)}^{v(x)} f(t)dt = f(v(x))v'(x) - f(u(x))u'(x).$$

This formula assumes the integrand is a function of t alone. If the integrand also contains x, account for that dependence separately.

Example 24. If $A(x) = \int_{x^2}^{x^3}(1 + t^2)dt$, then $A'(x) = 3x^2(1 + x^6) - 2x(1 + x^4)$.

Example 25. Let $B(x) = \int_0^x (x - t)(t + 2)dt$. Rewrite it as $x\int_0^x (t + 2)dt - \int_0^x t(t + 2)dt$. Differentiate using the product rule; the endpoint terms cancel, leaving $B'(x) = \int_0^x (t + 2)dt = x^2/2 + 2x$. Thus $B''(x) = x + 2$.

A continuous nonnegative integrand makes A nondecreasing. A strictly positive one makes it strictly increasing. Zeros and sign changes of f determine the stationary points of A.

12 Limits sums recurrences and inverse functions

To recognise a Riemann sum, isolate a width such as $1/n$ and rewrite the remaining expression as $f(r/n)$. Do this before taking the limit.

Example 26. For

$$L = \lim_{n\to\infty} \sum_{r=1}^{n} \frac{2n}{4n^2 + r^2},$$

write the term as $(1/n)\,2/[4 + (r/n)^2]$. Therefore $L = \int_0^1 2/(4 + x^2)dx = \tan^{-1}(1/2)$.

A recurrence. Let $S_n = \int_0^{\pi/2} \sin^n x\,dx$, for integers $n \geq 0$. For $n \geq 2$, integration by parts with $u = \sin^{n-1} x$ and $dv = \sin x\,dx$ gives

$$S_n = (n-1)\int_0^{\pi/2} \sin^{n-2} x \cos^2 x\,dx = (n-1)(S_{n-2} - S_n).$$

Hence $S_n = (n-1)S_{n-2}/n$, with $S_0 = \pi/2$ and $S_1 = 1$. For example, $S_6 = (5/6)(3/4)(1/2)(\pi/2) = 5\pi/32$. The boundary product vanishes for $n \geq 2$.

An inverse-function identity. If f is continuously differentiable and strictly increasing on $[a, b]$, substitute $y = f(x)$ and integrate $xf'(x)$ by parts:

$$\int_{f(a)}^{f(b)} f^{-1}(y)dy = bf(b) - af(a) - \int_a^b f(x)dx.$$

An explicit formula for the inverse is unnecessary. For $f(x) = x^3 + 2x$ on $[0, 2]$, the integral of the inverse from 0 to 12 is $24 - \int_0^2 (x^3 + 2x)dx = 24 - 8 = 16$.

13 Bounds and parameter reasoning

If $m \leq f(x) \leq M$ on $[a, b]$, then $m(b-a) \leq \int_a^b f \leq M(b-a)$. Integrating a pointwise inequality is often enough to reject an impossible answer. Also,

$$\left|\int_a^b f\right| \leq \int_a^b |f|, \qquad \left(\int_a^b fg\right)^2 \leq \left(\int_a^b f^2\right)\left(\int_a^b g^2\right).$$

The second inequality is the integral form of Cauchy–Schwarz, valid for square-integrable real functions. For continuous functions, it follows by integrating $(f - \lambda g)^2 \geq 0$ and considering the resulting quadratic in λ.

Example 27. Without integrating, $1/8 < \int_0^1 dx/(7 + x^6) < 1/7$, since the integrand lies strictly between these constants for $0 < x < 1$.

Example 28. The parameter a minimizing $\int_0^4 (x - a)^2 dx$ is found by completing the square after integration:

$$\int_0^4 (x-a)^2 dx = 4a^2 - 16a + 64/3 = 4(a-2)^2 + 16/3.$$

Thus $a = 2$ and the minimum is $16/3$. This is also the midpoint, the average value of x over a uniform interval.

Beyond the core route. Improper endpoint limits and differentiation with respect to a parameter require additional existence and interchange arguments. Do not apply them merely because a symbolic expression looks familiar. The advanced papers mainly use substitution, parts, symmetry and the fundamental theorem, so every solution remains traceable to these notes.

14 A reliable problem solving routine

1. Read the domain and limits. Mark poles, zeros, and formula-change points.
2. Simplify algebraically before selecting a method. A division or identity may remove most of the work.
3. Test substitution, standard quadratic forms, parts, or partial fractions. For a definite integral also test reflection, parity and periodicity.
4. Write the actual transformed differential and limits, or the parts choices. Carry signs and factors explicitly.
5. Finish with an exact answer and the integration constant when appropriate. Check by differentiation, sign, size, or an independent symmetry argument.

Frequent mistakes to avoid. Omitting C; applying the power rule to $1/x$; forgetting an inner derivative; treating $\sqrt{u^2}$ as u without a sign condition; keeping old bounds after substitution; assuming oddness on an asymmetric interval; confusing signed integral with total area; and subtracting antiderivative values across a singularity.

Suggested study route. First reproduce Examples 1–16 and attempt Practice 1–25. Next work through Examples 17–23 and Practice 26–38. Then study the remaining examples and Practice 39–50. Use a timed paper only after correcting practice errors; keep a record of whether each error involved method selection, algebra, signs or interpretation.

Scope and further reading

These notes cover indefinite and definite integrals. Geometric area is introduced only to explain signed accumulation; a full treatment of areas between curves belongs to the separate Applications of Integrals chapter. The more demanding sections extend the core chapter without labelling students by examination category.

The official CBSE Mathematics 041 curriculum for 2026–27 includes substitution, parts, partial fractions, standard algebraic forms, the fundamental theorem and definite-integral properties. Curriculum reference: `https://cbseacademic.nic.in/web_material/CurriculumMain27/SecPart2/Maths_SecP2_2026-27.pdf`. Explanations and examples in this document were newly written for Ramavanamath.com.